Environmental Science (DCE-201)

UNIT II

Pollution

Pollution is the undesirable change in physical, chemical, or biological characteristics of the environment that harms living organisms and natural resources.

It can affect:

- Air
- Water
- Soil
- Sound environment (noise)

Pollutant

A **pollutant** is any substance (solid, liquid, gas, or energy like heat/sound) that causes pollution.

Types of Pollutants

1. **Primary Pollutants** – Emitted directly
   Examples: CO, SO_2, NO, smoke, dust
2. **Secondary Pollutants** – Formed by reactions in the atmosphere
   Examples: Ozone (O_3), PAN, acid rain

Air Pollution

Air pollution is the presence of harmful substances (gases, dust, smoke, fumes, or vapors) in the atmosphere in concentrations that are harmful to humans, animals, plants, or materials.

Major Air Pollutants

- Carbon Monoxide (CO)
- Sulphur Dioxide (SO_2)
- Nitrogen Oxides (NO_x)
- Particulate Matter (PM10, PM2.5)
- Hydrocarbons (HC)
- Ozone (O_3)

Sources of Air Pollution

A. Natural Sources

Source	Pollutants Produced
Volcanic eruptions	SO_2, ash
Forest fires	CO, CO_2, smoke
Dust storms	Particulate matter
Sea spray	Salt particles
Pollen & microbes	Biological particles

B. Man-Made (Anthropogenic) Sources

1. Refrigerants

Used in ACs and refrigerators.

Pollutants Released:

- CFCs (Chlorofluorocarbons)
- HCFCs
- HFCs

Effects:

- Ozone layer depletion (CFCs)
- Global warming (HFCs)

2. I.C. Engines (Internal Combustion Engines)

Found in cars, bikes, trucks, generators.

Pollutants Emitted:

- CO (incomplete combustion)
- NO_x (high-temperature combustion)
- Hydrocarbons (unburnt fuel)
- Particulate matter (diesel engines)
- CO_2 (greenhouse gas)

3. Boilers (Industrial & Thermal Power Plants)

Boilers burn coal, oil, or gas to produce steam.

Pollutants Released:

- SO_2 (from sulfur in coal)
- NO_x
- Fly ash (fine particles)
- CO and CO_2

Major Impact:

Industrial smog, acid rain, respiratory diseases

Noise Pollution

Noise pollution is unwanted or excessive sound that disturbs normal activities and harms human and animal health.

Common Sources of Noise Pollution

Source	Examples
Transportation	Road traffic, trains, aircraft
Industrial	Machinery, generators, compressors
Construction	Drilling, hammering
Domestic	Loudspeakers, TV, music systems

Effects of Noise Pollution

- Hearing loss
- Stress and anxiety
- Sleep disturbance
- High blood pressure
- Reduced concentration

AIR POLLUTANTS – TYPES

Air pollutants are unwanted solid, liquid, or gaseous substances present in the atmosphere in harmful concentrations.

1. Classification Based on Origin

Primary Pollutants

Emitted directly from sources.

- Carbon Monoxide (CO) – vehicles
- Sulfur Dioxide (SO_2) – coal burning

- Nitrogen Oxides (NO_x) – automobiles, power plants
- Particulate Matter (PM) – dust, smoke
- Volatile Organic Compounds (VOCs) – paints, solvents

Secondary Pollutants

Formed by chemical reactions in the atmosphere.

- Ozone (O_3)
- Peroxyacetyl Nitrate (PAN)
- Secondary particulate matter
- Photochemical smog

Classification Based on Physical State

Gaseous Pollutants

SO_2, NO_x, CO, CO_2, O_3, NH_3, VOCs

Particulate Pollutants

Tiny solid or liquid particles suspended in air.

PARTICULATE POLLUTANTS (PM)

Particulate matter consists of finely divided solids or liquids.

Types Based on Size

Type	Size	Examples
PM10	< 10 µm	Dust, pollen, mold
PM2.5	< 2.5 µm	Smoke, combustion particles
Ultrafine	< 0.1 µm	Vehicle exhaust particles

Sources of Particulate Matter

Natural Sources

- Volcanic eruptions
- Forest fires
- Dust storms
- Sea spray

Man-made Sources

- Thermal power plants

- Cement industries
- Brick kilns
- Diesel vehicles
- Construction activities

EFFECTS OF PARTICULATE POLLUTANTS

1. Health Effects

- Irritation of eyes, nose, throat
- Asthma and bronchitis
- Reduced lung function
- Heart diseases
- Lung cancer (long-term PM2.5 exposure)

2. Environmental Effects

- Reduced visibility (haze)
- Acid rain formation
- Damage to crops and vegetation
- Soil and water contamination

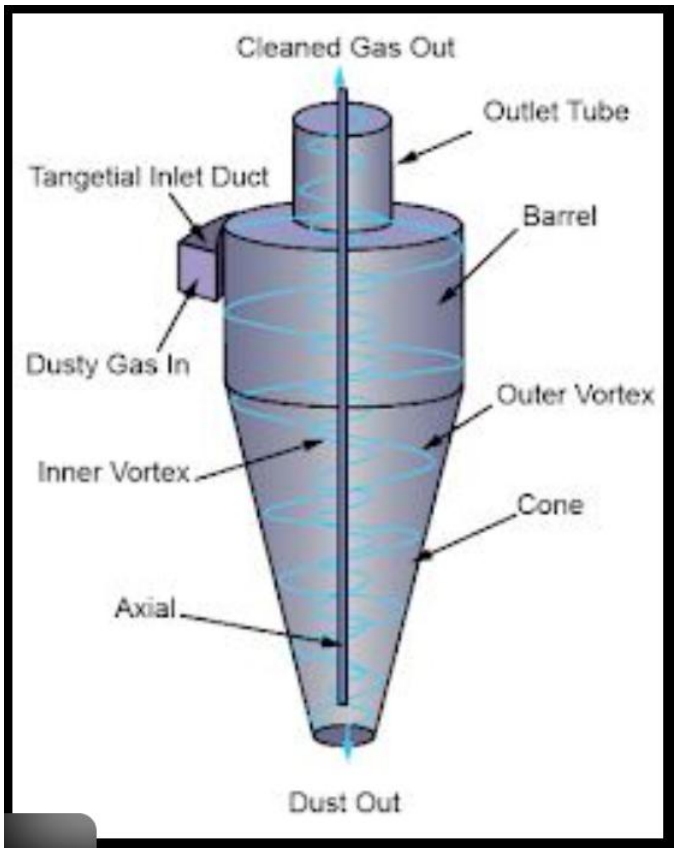
3. Material Damage

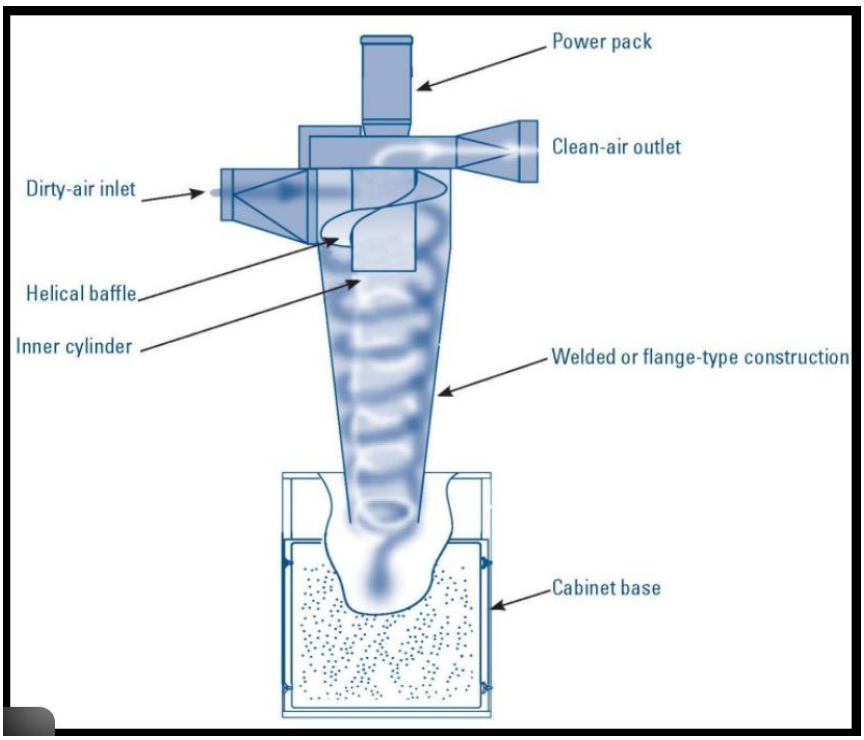
- Corrosion of metals
- Soiling of buildings and monuments

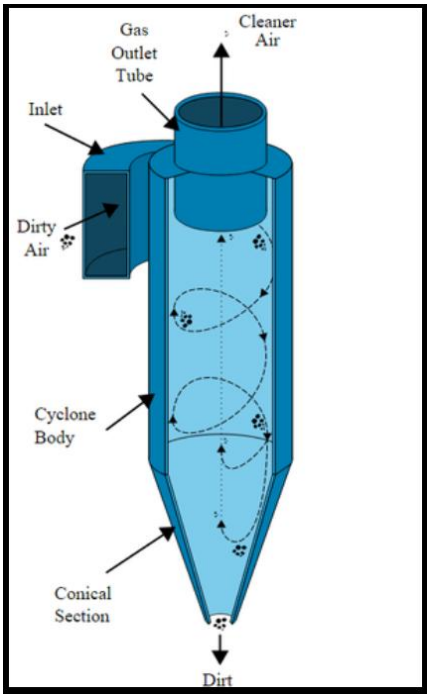
CONTROL OF PARTICULATE POLLUTANTS

Control devices remove dust from industrial exhaust gases.

Cyclone Separator







Principle

Works on **centrifugal force**.

Working

- Polluted gas enters tangentially
- Spiral motion is created
- Heavier particles move outward and fall into a hopper
- Clean gas exits from the top

Efficiency

- Good for particles **>10 µm**
- Not effective for very fine particles

Advantages

✓ Simple construction

✓ Low cost

✓ Low maintenance

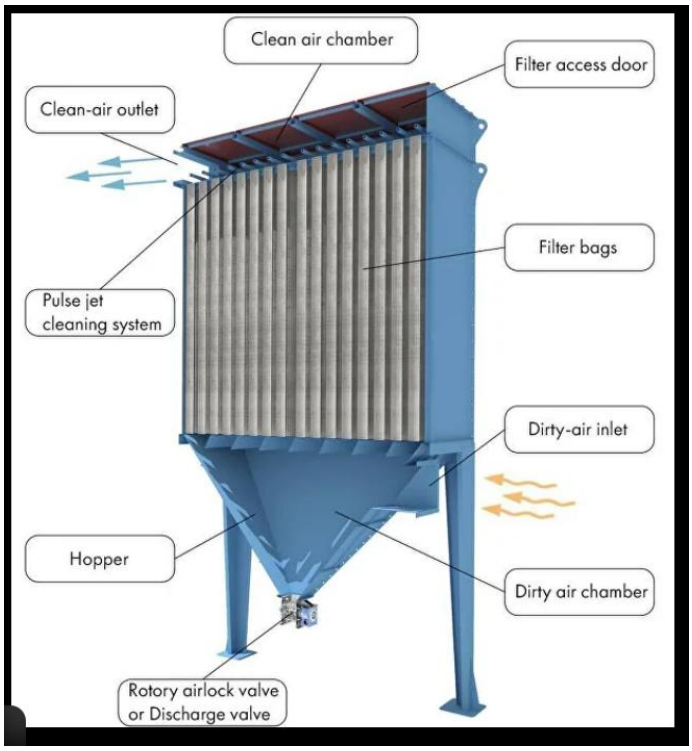
✓ Handles high temperature gases

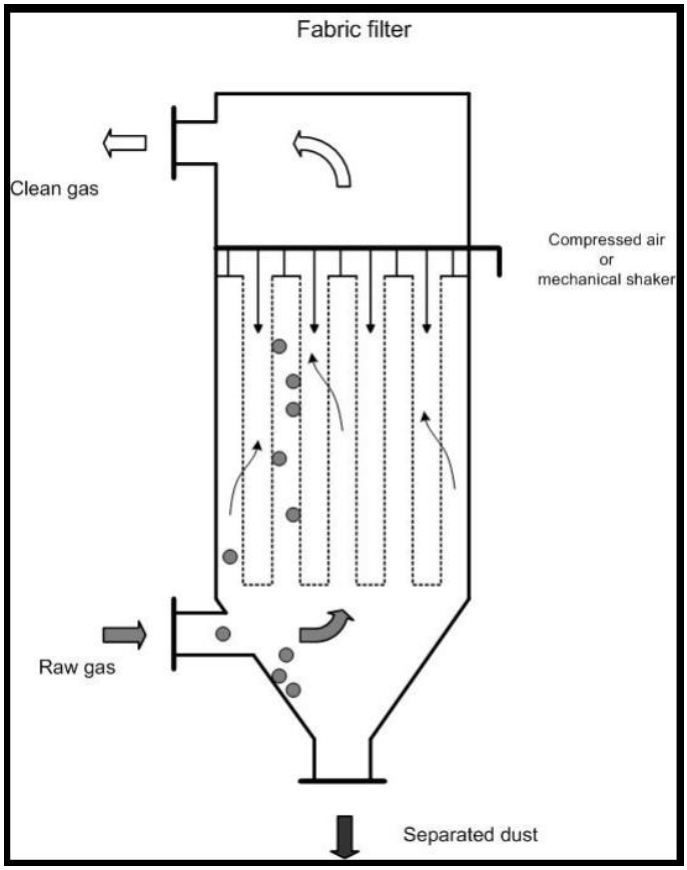
Disadvantages

✕ Inefficient for PM2.5

✕ Pressure drop

Bag Filter (Fabric Filter)





Principle

Filtration through fabric bags.

Working

- Dust-laden gas passes through fabric bags
- Particles get trapped on fabric surface
- Clean gas exits
- Dust cake removed by shaking or reverse air pulse

Efficiency

- Very high (up to **99%**)
- Effective for **fine particles**

Advantages

✓ High collection efficiency

✓ Works for small particle sizes

✓ Flexible operation

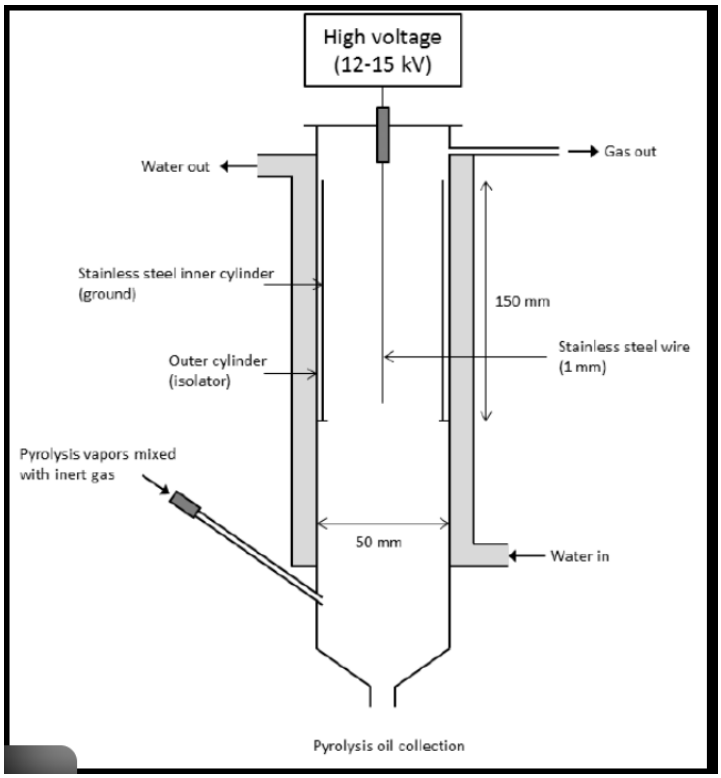
Disadvantages

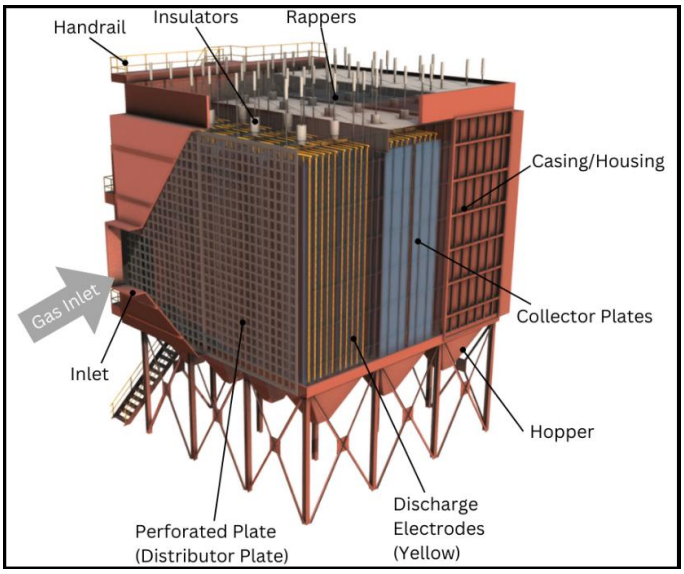
✕ Not suitable for very high temperatures

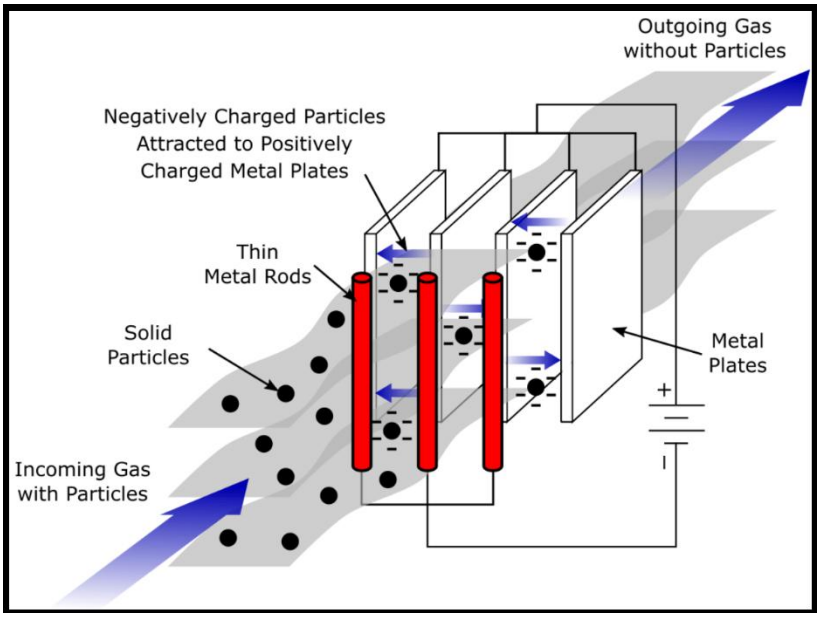
✕ Fabric damage possible

✕ Maintenance required

Electrostatic Precipitator (ESP)







Principle

Uses **electrostatic force** to remove particles.

Working

1. Gas passes through ionization chamber
2. Particles receive negative charge
3. Charged particles move toward positive collecting plates
4. Dust removed by rapping mechanism

Efficiency

- Very high (up to **99.9%**)
- Effective for very fine particles

Advantages

✓ Low pressure drop

✓ Handles large gas volumes

✓ Suitable for high temperatures

Disadvantages

✕ High initial cost

✕ Requires skilled maintenance

✕ Performance affected by particle resistivity

COMPARISON OF DEVICES

Device	Principle	Particle Size Removed	Efficiency	Cost
Cyclone	Centrifugal force	Coarse (>10 µm)	Medium	Low
Bag Filter	Filtration	Fine particles	Very High	Medium
ESP	Electrostatic attraction	Very fine particles	Extremely High	High

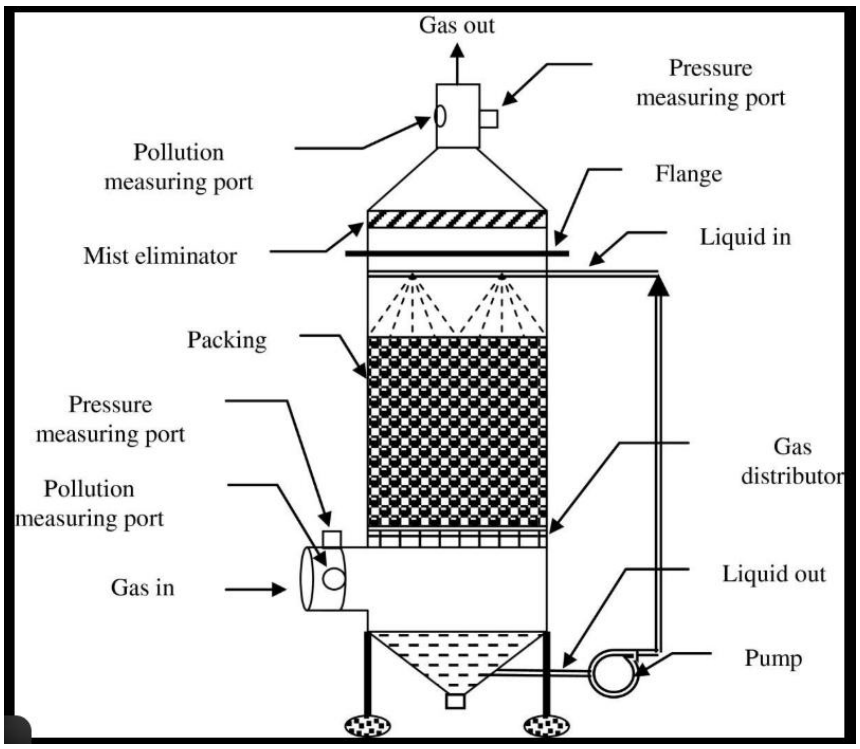
Gaseous Pollution Control

Gaseous pollution control involves removal of harmful gases like **SO_2, NO_x, CO, VOCs, H_2S, NH_3** from industrial and vehicular emissions before release into the atmosphere.

Methods of Control

1. Absorption
2. Adsorption
3. Condensation
4. Combustion
5. Catalytic conversion

Absorber (Gas Absorption)



An absorber is a device in which gaseous pollutants are removed by dissolving them into a liquid solvent.

Principle

Based on **mass transfer** between gas and liquid phase.

Construction

- Vertical column
- Packing material (Raschig rings, saddles)
- Gas inlet at bottom
- Liquid inlet at top

Working

- Polluted gas enters from bottom
- Liquid solvent flows downward
- Pollutant transfers from gas to liquid

Examples

- SO_2 absorbed in lime slurry
- NH_3 absorbed in water

Advantages

- High removal efficiency
- Suitable for large gas volumes

Disadvantages

- Produces wastewater
- Corrosion issues

Catalytic Converter

A catalytic converter is an emission control device used in automobiles to convert harmful gases into harmless gases using catalysts.

Location

Installed in vehicle exhaust system.

Catalyst Used

- Platinum (Pt)
- Palladium (Pd)
- Rhodium (Rh)

Reactions (Three-way Converter)

1. $CO \rightarrow CO_2$

2. Hydrocarbons → $CO_2 + H_2O$
3. $NO_x \rightarrow N_2 + O_2$

Importance

- Reduces air pollution
- Mandatory under BS-VI norms

Limitations

- Expensive
- Catalyst poisoning by lead

Effects of Air Pollution due to Refrigerants

Common Refrigerants

- CFCs
- HCFCs
- HFCs

Effects

1. Ozone Layer Depletion

CFCs release chlorine radicals that destroy ozone.

2. Global Warming

Refrigerants have high Global Warming Potential (GWP).

3. Health Effects

- Skin cancer
- Cataracts
- Weakened immunity

4. Environmental Effects

- Climate change
- Rising sea levels

Control Measures

- Use eco-friendly refrigerants (R-134a, R-290)
- Proper leakage control
- Compliance with Montreal Protocol

Air Pollution due to I.C. Engines

Pollutants Emitted

- CO

- NO_x
- Hydrocarbons
- SO_2
- Particulate matter
- CO_2

Causes

- Incomplete combustion
- Poor engine maintenance
- High combustion temperature

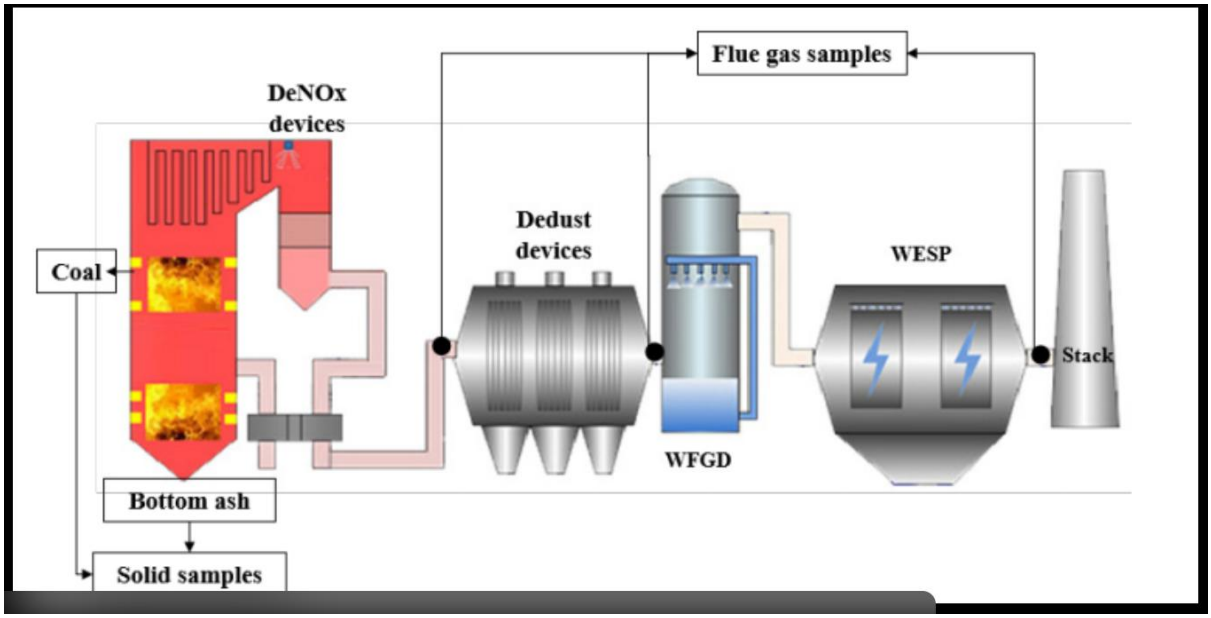
Effects

- Smog formation
- Acid rain
- Respiratory diseases
- Global warming

Control Measures

- Catalytic converters
- Cleaner fuels (CNG, LPG)
- Electric vehicles
- Proper engine tuning

Air Pollution due to Boilers



Sources

- Coal-fired boilers
- Oil-fired boilers
- Thermal power plants

Pollutants

- SO_2
- NO_x
- CO
- Fly ash
- CO_2

Effects

- Acid rain
- Lung diseases
- Visibility reduction
- Climate change

Control Devices

- Electrostatic precipitator (ESP)
- Bag filter
- Wet scrubber
- Flue Gas Desulfurization (FGD)

Noise Pollution

Noise pollution means unwanted or excessive sound that disturbs people and the environment. When sound becomes harmful or uncomfortable, it is called noise. It is a common problem in cities due to increasing vehicles, industries, and population.

Sources of Noise Pollution

Noise pollution is mainly caused by human activities. Road traffic, vehicle horns, trains, and airplanes are major sources. Industries and factories produce noise from heavy machines and generators. Construction activities such as drilling and demolition also create loud sounds. Household appliances, loudspeakers, and firecrackers add to the problem, especially during public events and festivals.

Measurement of Noise

Noise is measured in **decibels (dB)** using a **Sound Level Meter**. The dB(A) scale is used because it matches human hearing.

A whisper is about 30 dB, normal conversation is around 60 dB, traffic noise ranges from 70–90 dB, and aircraft take-off can reach 120 dB or more. Continuous exposure above 85 dB can damage hearing.

Effects of Noise Pollution

Noise pollution affects human health by causing hearing loss, stress, headache, high blood pressure, and sleep disturbance. It reduces concentration and work efficiency. In children, it can affect learning ability.

It also disturbs animals by affecting their communication, breeding, and migration. Overall, it lowers the quality of life.

Noise Pollution (Regulation and Control) Rules, 2000

In India, noise pollution is controlled under these rules, framed under the Environment (Protection) Act, 1986.

The rules fix permissible noise limits for industrial, commercial, residential, and silence zones. Lower limits are set for night time. Loudspeakers require permission and are generally banned between 10 PM and 6 AM. Silence zones include areas near hospitals, schools, and courts.

Prevention and Control

Noise pollution can be reduced by avoiding unnecessary honking, maintaining vehicles, keeping sound systems at low volume, planting trees, and using soundproof materials. Strict enforcement of laws and public awareness are important for effective control.

Noise pollution is a serious environmental issue that affects health and the environment. Proper regulation, public cooperation, and awareness are essential to reduce its impact.